Infection and the Cholestatic Infant: Cytomegalovirus, Syphilis, Urinary Infection, Toxoplasma, and the Viruses That Cause Neonatal Liver Failure.

Imagine three infants on the same ward on the same day. The first is a five-week-old boy who is feeding well and gaining weight, but whose stools have become pale. His urine has just come back positive for cytomegalovirus by polymerase chain reaction, a DNA amplification test. The second is a three-week-old girl who has been vomiting, is a little sleepy, and whose urine culture is growing Escherichia coli. The third is a ten-day-old with a fever, no rash at all, an alanine aminotransferase in the thousands and blood that will not clot. All three have an infection, and all three have a liver problem. Yet the right action is completely different in each: in the first the infection may be a distraction from biliary atresia, in the second the infection is the cure-able cause but may also be a clue to galactosemia, and in the third the infection will kill within days unless an antiviral is started before any result is back.

The clinical question of this chapter is how infection causes cholestasis and liver failure in the first months of life, and how to sort the infectious causes by urgency. The main idea is this. Infection produces two very different livers. Bacterial infection, including urinary infection and sepsis, causes a functional and reversible cholestasis: the liver cells are alive, but the machinery that pumps bile out of them has been switched down by inflammation, and it switches back on when the infection is treated. Herpes simplex virus and the enteroviruses do something else. They kill liver cells directly, causing massive necrosis and acute liver failure, often in an infant whose skin shows nothing. Between these two extremes sit the congenital infections, cytomegalovirus, syphilis and toxoplasma, each of which has its own diagnostic trap and its own treatment rules.

We will move through the topic in order. First, the mechanism of inflammatory cholestasis. Second, how often infection explains a cholestatic infant. Third, urinary infection. Fourth, congenital cytomegalovirus, its diagnosis, its treatment, and its uneasy relationship with biliary atresia. Fifth, congenital syphilis and toxoplasmosis. Sixth, neonatal acute liver failure, with herpes simplex virus and the enteroviruses. We then compare the North American and European documents, list the common traps, and finish with what to do differently in clinic.

A note on sources. The central guideline is the 2017 guideline on the evaluation of cholestatic jaundice in infants, written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. For frequency we use the 2015 systematic review by Gottesman and colleagues in BMC Pediatrics, which pooled 1,692 infants with conjugated hyperbilirubinemia. For cytomegalovirus we use the randomized trials of Kimberlin and colleagues, published in 2003 and 2015, and the 2017 international consensus led by Rawlinson in the Lancet Infectious Diseases. For syphilis we use the surveillance data and the 2021 treatment guidelines of the United States Centers for Disease Control and Prevention, the CDC. For liver failure we use the young infant cohort of the Pediatric Acute Liver Failure Study Group, published by Sundaram and colleagues in 2011, together with the neonatal

herpes trials and the 2023 World Health Organization outbreak reports on echovirus 11.

Part one. How inflammation switches off the bile pump.

To understand why a urinary infection can make an infant yellow, start with how bile is made. Bile acids travel from the portal blood into the hepatocyte through uptake transporters on the side of the cell that faces the blood, mainly the sodium taurocholate cotransporting polypeptide, called NTCP, and the organic anion transporting polypeptides, called OATPs. They leave the cell into the bile canaliculus through export pumps on the opposite side, mainly the bile salt export pump, called BSEP, and, for conjugated bilirubin, the multidrug resistance associated protein 2, called MRP2. Bile flow depends on these pumps working at full capacity.

Now add a Gram-negative infection. Lipopolysaccharide, also called endotoxin, from the bacterial wall binds a receptor called toll-like receptor 4 on Kupffer cells, the resident macrophages of the liver. The Kupffer cells release inflammatory cytokines, chiefly tumor necrosis factor alpha, interleukin 1 beta and interleukin 6. These cytokines act on the hepatocyte nucleus and reduce the transcription of the transporter genes. Uptake through NTCP and the OATPs falls. Export through BSEP and MRP2 falls, and MRP2 is also pulled back from the canalicular membrane into the cell. The hepatocyte protects itself by increasing two escape pumps on the blood side, MRP3 and MRP4, which push conjugated bilirubin and bile acids back into the circulation. The net result is a rise in conjugated bilirubin in the blood.

The classic experiment, published by Trauner and colleagues in the Journal of Clinical Investigation in 1998, showed that after a single dose of endotoxin in rats, the messenger RNA for the uptake transporter reached its lowest point at about sixteen hours, because the transcription factors that drive it lost about half of their binding activity, and it had recovered by forty-eight hours. That time course is the key teaching point. Inflammatory cholestasis is a regulated, reversible shutdown, not cell death. Clinically, the direct bilirubin rises, the aminotransferases are only modestly raised, and once the infection is treated the bilirubin falls over days to weeks.

Viral hepatitis in the newborn is different in kind. Herpes simplex virus and the enteroviruses infect and kill hepatocytes. In the young infant cohort of the Pediatric Acute Liver Failure Study Group, the median alanine aminotransferase in viral liver failure was 618 units per liter and the median aspartate aminotransferase was 2,863, whereas in gestational alloimmune liver disease, the condition formerly called neonatal hemochromatosis, the medians were only 35 and 84. So a very high transaminase in a sick neonate points toward a virus, while liver failure with near normal transaminases points toward an alloimmune or metabolic cause.

Part two. How often is it infection?

The Gottesman review gives a useful sense of proportion. Among 1,692 infants with conjugated hyperbilirubinemia, infection was the final diagnosis in 194, which is about eleven and a half percent, or roughly one infant in nine. Within those 194 infections, cytomegalovirus was the largest group at about one third, thirty-three and a half percent. Bacterial sepsis came next at about one quarter, twenty-four point seven percent. Congenital syphilis was about eleven percent, E. coli

urinary infection almost ten percent, rubella about six percent, toxoplasmosis between three and four percent, hepatitis B under two percent and herpes simplex virus only one percent.

Two lessons follow. First, if we add sepsis, urinary infection and syphilis together, bacterial causes make up nearly half of all infectious cholestasis, and every one of them is curable within days. Second, herpes simplex virus is almost absent from cholestasis series. That is not because it rarely affects the liver. It is because it presents as acute liver failure in the first two weeks of life, not as prolonged jaundice in a well infant, so it is counted in liver failure registries rather than in cholestasis reviews.

Part three. The well-looking infant with bacteriuria.

Urinary infection is the infection most easily missed, because the infant may have no fever and may look well. In a study published in Pediatrics in 2002, Garcia and Nager found urinary infection in seven and a half percent of afebrile, otherwise asymptomatic jaundiced infants younger than eight weeks, and infection was more likely when the conjugated fraction of bilirubin was raised. In a Turkish study published in 2006, Bilgen and colleagues found urinary infection, proven on catheter samples, in eight of 102 well term newborns with raised indirect bilirubin.

But the numbers depend heavily on where and how one looks. A 2016 local review and meta-analysis in the Archives of Disease in Childhood by Steadman and colleagues found no confirmed infection at all among 279 well infants with prolonged jaundice in their own British unit. The pooled prevalence in British studies was about two tenths of one percent, compared with about eight percent in studies from the rest of the world. Why such a gap? Bag urine specimens are easily contaminated, loose colony count thresholds count contaminants as infection, and referral populations are selected for sicker infants. When strict criteria are used, a catheter sample with pyuria and a single organism, the prevalence falls.

What do the documents say? The 2017 joint NASPGHAN and ESPGHAN guideline states that a urine culture should be obtained early in the diagnostic evaluation of a cholestatic infant. The 2004 hyperbilirubinemia guideline of the American Academy of Pediatrics, the AAP, asked for a urinalysis and urine culture when the direct bilirubin was raised. Its 2022 revision no longer mentions urine testing at all; it lists urinary infection only as one of the causes of direct hyperbilirubinemia. The practical answer for a hepatology service is simple. In an infant with conjugated hyperbilirubinemia, send a catheter urine for culture. It is cheap, it may reveal a curable cause, and, as we saw in the previous lesson, an E. coli infection in a cholestatic infant is also a classic clue to galactosemia, so lactose should be stopped until galactosemia is excluded.

Part four. Congenital cytomegalovirus: the commonest infection, the hardest to date.

Cytomegalovirus infects about one to two percent of newborns according to the 2017 guideline, and the 2024 consensus of the European Congenital Infection Initiative puts the global birth prevalence at about two thirds of one percent. Most infected newborns have no symptoms. Among the symptomatic infants enrolled in the 2015 valganciclovir trial, between forty-five and fifty-one

percent had hepatitis at entry, meaning raised aminotransferases or bilirubin, and between forty-three and fifty-five percent had an enlarged liver. The histology, when a biopsy is done, may show giant cell hepatitis, inflammation of the bile ducts and extramedullary blood formation.

The central diagnostic problem is dating the infection. Congenital infection, acquired in the womb, can only be proven from a sample taken in the first three weeks of life, that is, within twenty-one days of birth. After that, a positive urine test may reflect postnatal infection acquired from breast milk or from a blood transfusion, which in a term infant is usually harmless. Cholestatic infants typically reach a hepatologist at four to eight weeks of age, so a positive urine test at six weeks does not prove congenital infection. The fix is retrospective. Most newborn screening programs store the dried blood spot card, and polymerase chain reaction for cytomegalovirus on that stored spot can date the infection. A positive result proves congenital infection. A negative result does not exclude it, because the dried blood spot test has limited sensitivity. Saliva testing is sensitive and convenient, but in a breastfed infant saliva can be contaminated by virus in the milk, so a positive saliva result should be confirmed in urine. The 2017 guideline adds that IgM antibody is of limited diagnostic value, and that urine polymerase chain reaction is the test of choice.

Now treatment. In 2003 Kimberlin and colleagues published a randomized trial of six weeks of intravenous ganciclovir in infants with symptomatic congenital infection involving the central nervous system. At six months, hearing had worsened in none of the treated infants compared with forty-one percent of the untreated infants. The price was toxicity: grade three or four neutropenia, a dangerous fall in neutrophils, occurred in sixty-three percent of treated infants.

The 2015 trial in the New England Journal of Medicine then compared six months against six weeks of oral valganciclovir, the oral prodrug of ganciclovir, at sixteen milligrams per kilogram twice daily, in 86 evaluable infants with symptomatic disease. At six months, hearing in the better ear did not differ between the groups. But hearing across both ears was improved or normal in seventy-three percent of the six month group versus fifty-seven percent of the six week group at twelve months, and seventy-seven percent versus sixty-four percent at twenty-four months. Language scores on the Bayley developmental test at two years also favored the longer course. Severe neutropenia after the first six weeks did not differ significantly between the arms.

Who should be treated? The 2017 international consensus recommends valganciclovir for six months for infants with moderate to severe symptomatic disease, starting within the first month of life. It does not recommend routine treatment for asymptomatic infants or for mildly symptomatic infants, meaning those with one or two mild and transient findings, such as a single raised aminotransferase or slight liver enlargement. Hepatitis was accepted as a sign of symptomatic disease for entry into the 2015 trial, but no trial has shown that antiviral therapy changes the course of the liver disease itself. In practice, therefore, the decision to treat is driven by the brain, the hearing and the overall severity, not by the transaminases.

Part five. Cytomegalovirus and biliary atresia.

There is a subgroup of infants with biliary atresia who also test positive for cytomegalovirus IgM, and this has raised two questions. Does the virus cause atresia, and does it change the outcome? On

outcome, the data are fairly consistent. In a 2015 series of 129 infants from King's College Hospital, published by Zani and colleagues, twenty infants were IgM positive. Only fifteen percent of them cleared their jaundice after the Kasai operation, compared with fifty-two percent of the IgM negative infants. A 2021 meta-analysis by Zhao and colleagues pooled eight studies and found that cytomegalovirus positivity roughly halved the odds of clearing jaundice, with an odds ratio of 0.47. Native liver survival, however, did not differ significantly, with a hazard ratio of 1.59 and a confidence interval that crossed one. The proportion of atresia infants who are virus positive varies by region, from about ten to thirty-eight percent in series from England, Germany, Brazil and Sweden, and twenty-seven percent in a 2022 West China cohort of 175 infants.

On causation, the 2017 guideline is clear: a causal role for cytomegalovirus in biliary atresia remains unproven. The practical message is the important one. A positive cytomegalovirus test in a cholestatic infant with pale stools never explains away the pale stools. The infant still needs the full atresia work-up, including biopsy and cholangiography, on the usual urgent schedule.

Part six. Congenital syphilis: a re-emerging cause of neonatal hepatitis.

In the United States, congenital syphilis has returned on a scale not seen in decades. CDC surveillance recorded 334 cases in 2012, 1,306 in 2018, 1,884 in 2019, 2,163 in 2020, 2,881 in 2021, 3,769 in 2022 and 3,882 in 2023, a rate of almost 106 cases per 100,000 live births. That is close to a twelvefold rise in eleven years, and provisional data for 2024 show nearly 4,000 cases. In a 2023 South African cohort of fifty symptomatic neonates, eighty percent had an enlarged liver and spleen, and the CDC guideline lists conjugated hyperbilirubinemia and cholestatic jaundice among the examination findings of congenital syphilis. The liver disease is an interstitial hepatitis with fibrosis.

The 2021 CDC treatment guidelines sort every infant born to a mother with reactive serology into four scenarios. In scenario one, proven or highly probable disease, the infant has an abnormal examination, or a nontreponemal antibody titer at least four times the maternal titer, or a positive darkfield or polymerase chain reaction test. These infants receive aqueous crystalline penicillin G, fifty thousand units per kilogram intravenously every twelve hours for the first seven days of life and every eight hours thereafter, for ten days in total. If more than one day of therapy is missed, the whole course is restarted. In scenario two, possible disease, the examination is normal but the mother was untreated, inadequately treated, or treated less than four weeks before delivery; these infants receive the same ten-day course, or a single dose of benzathine penicillin if the full evaluation is normal and follow-up is certain. Scenario three, less likely disease, is a mother adequately treated during pregnancy more than four weeks before delivery, and the infant receives a single dose of benzathine penicillin. Scenario four, unlikely disease, is a mother adequately treated before pregnancy with a low and stable titer, and the infant needs only serologic follow-up. The key point for the hepatologist is simple: a cholestatic infant with congenital syphilis has an abnormal examination and is therefore scenario one by definition. Always ask for the mother's treponemal and nontreponemal results, and when and how she was treated.

Part seven. Toxoplasmosis: rare in the liver clinic, long in treatment.

Congenital toxoplasmosis affects about one and a half per thousand live births worldwide, according to the 2013 estimate of Torgerson and Mastroiacovo in the Bulletin of the World Health Organization, but it accounted for only three to four percent of infectious cholestasis in the Gottesman review, seven infants in all. Most infected newborns look normal. The clues are outside the liver: chorioretinitis, hydrocephalus, intracranial calcification, and a history of maternal seroconversion in pregnancy. Generalized disease can bring an enlarged liver and spleen and jaundice. Diagnosis rests on infant IgM and IgA antibodies, on comparing the infant's IgG over time with the mother's, and on polymerase chain reaction of blood, urine or cerebrospinal fluid, together with an eye examination and cranial imaging. Treatment is pyrimethamine, sulfadiazine and folinic acid for twelve months, with regular blood counts because neutropenia is common, and steroids only for active eye disease or a very high spinal fluid protein. The 2017 guideline does not recommend routine serology for the so-called TORCH infections. Test for toxoplasma when the eye, the brain or the maternal history suggests it, not as a reflex panel.

Part eight. Neonatal acute liver failure: where the viruses sit.

Now we move from cholestasis to liver failure. The young infant cohort of the Pediatric Acute Liver Failure Study Group included 148 infants aged ninety days or younger, with a median age of eighteen days. The cause was indeterminate in about thirty-eight percent, metabolic disease in about nineteen percent, gestational alloimmune liver disease in about thirteen and a half percent, herpes simplex virus in about thirteen percent, enterovirus in under three percent and cytomegalovirus in under one percent. At twenty-one days, sixty percent had survived with their own liver, sixteen percent had been transplanted and twenty-four percent had died. Viral liver failure was by far the most lethal group: sixty-four percent died without a transplant, compared with twenty-nine percent in metabolic disease, sixteen percent in alloimmune disease and fourteen percent in indeterminate failure. Not one of the twenty-four viral cases was transplanted, and twenty were never listed, because these infants have overwhelming multi-organ infection.

Part nine. Neonatal herpes simplex virus: liver failure without vesicles.

Neonatal herpes is uncommon but catastrophic. In a 2018 multicenter study in Pediatrics, Cruz and colleagues found herpes simplex virus in 0.42 percent of 26,533 infants aged sixty days or younger who had cerebrospinal fluid tested, and eighty percent of cases occurred in the first four weeks of life. In a 2016 series of forty-nine confirmed cases by Curfman and colleagues, forty-five percent had disseminated disease, thirty-three percent central nervous system disease and twenty percent disease limited to the skin, eyes and mouth. Sixteen percent presented with no seizure, no vesicles and no critical illness. Disseminated disease typically appears between days five and twelve, with fever or low temperature, lethargy, very high aminotransferases, coagulopathy and disseminated intravascular coagulation, and often in an infant whose mother never had a recognized lesion.

A 2025 report of the National Child Mortality Database in England makes the liver message stark. Thirty-five infants in England died of herpes simplex virus between April 2019 and December 2024. Twenty-three of them died within four days of presentation, and every one of those twenty-three had liver failure.

Dose and timing decide survival. In the trial of high dose acyclovir published by Kimberlin and colleagues in 2001, mortality in disseminated disease at twenty-four months was thirty-one percent with sixty milligrams per kilogram per day, compared with sixty-one percent with the older dose of thirty milligrams per kilogram per day, and historically about eighty-five percent without treatment. In central nervous system disease the corresponding figures were six percent, nineteen percent and roughly fifty percent. The current regimen is twenty milligrams per kilogram every eight hours, for twenty-one days in disseminated and central nervous system disease and fourteen days in disease limited to skin, eyes and mouth. A 2011 trial in the New England Journal of Medicine showed that six months of oral acyclovir suppression after central nervous system disease raised the mean Bayley mental development score at twelve months from sixty-eight to eighty-eight. And in a 2011 study of 1,086 American infants by Shah and colleagues, delayed acyclovir was associated with higher in-hospital mortality. The rule is therefore absolute. Any neonate with liver failure, or with sepsis and raised aminotransferases, receives intravenous acyclovir at once, with blood polymerase chain reaction, surface swabs and spinal fluid testing sent at the same time. Waiting for the result is the error that kills.

Part ten. Enteroviruses and parechovirus: the hemorrhage and hepatitis syndrome.

The enteroviruses, especially echovirus 11 and the coxsackie B viruses, can cause a neonatal syndrome of hepatic necrosis, coagulopathy and low platelets, often called the hemorrhage hepatitis syndrome. It usually begins in the first or second week of life with a sepsis-like illness, and the mother may have had a febrile illness around delivery. In 2023 the World Health Organization reported an unusual cluster of severe echovirus 11 infections in Europe. France recorded nine severe neonatal cases between July 2022 and April 2023, eight of them preterm and four of them twin pairs, and seven of the nine infants died. In 2024 Japan reported three neonates with liver failure from echovirus 11, all of whom died. In a Chinese series of 105 neonates with echovirus 11 infection in 2019, published in 2023, mortality was six percent overall and twenty percent among severe cases.

Treatment is mainly supportive. Intravenous immunoglobulin is widely used, but no randomized trial has shown that it improves survival. In a randomized trial of the antiviral pleconaril in sixty-one neonates, published by Abzug and colleagues in 2016, mortality was twenty-three percent with the drug versus forty-four percent with placebo, but pleconaril is not commercially available. Pocapavir, another capsid-binding drug, can be obtained only on compassionate use. Every neonate with liver failure should have enterovirus polymerase chain reaction sent on blood, spinal fluid and stool. A positive result changes counseling and infection control more than it changes therapy, and it usually means the infant is not a transplant candidate. Parechovirus, the close cousin of the enteroviruses, causes severe sepsis-like illness, seizures and meningoencephalitis in infants under three months, often with little or no inflammation in the spinal fluid, but liver involvement is uncommon.

Part eleven. What the guidelines say.

On urine culture in cholestasis, North America and Europe differ in an unusual way. The joint 2017 NASPGHAN and ESPGHAN guideline says to obtain one early, but the 2022 AAP guideline is silent, and British data suggest a very low yield in well infants. On routine TORCH serology they are aligned: both sides follow the joint guideline and recommend targeted polymerase chain reaction rather than a reflex antibody panel. On cytomegalovirus treatment they are aligned in the core: six months of valganciclovir for moderate to severe disease, with the 2024 European consensus adding prevention in pregnancy. On syphilis they are aligned: the CDC four scenario algorithm and the 2020 European guideline of the International Union against Sexually Transmitted Infections both use ten days of penicillin. The material difference is herpes simplex virus. American practice gives empiric acyclovir broadly to febrile neonates, whereas the 2021 neonatal infection guideline of the National Institute for Health and Care Excellence in England, known as NICE, does not include routine liver tests in the infection work-up, even though every early death in the English mortality review had liver failure. Finally, there is a gap. Neither the American Association for the Study of Liver Diseases nor the European Association for the Study of the Liver has a guideline on neonatal liver failure; the 2017 European liver failure guideline is written for adults and simply advises screening for viral causes.

Part twelve. Traps and pearls.

Four traps recur. The first is calling it infection: a positive cytomegalovirus test in an infant with pale stools stops the atresia work-up, or sepsis is blamed for the bilirubin while the stools are acholic. Infection and atresia can coexist. The second is missing the urine: a contaminated bag specimen is never repeated, antibiotics are started before the culture, or an E. coli infection is treated without a galactosemia screen. The third is dating cytomegalovirus late: a urine test at six weeks is called congenital, the newborn blood spot is never retrieved, or isolated transaminitis is treated with a toxic drug. The fourth is waiting on herpes: no vesicles, so no acyclovir; liver tests never sent in the septic neonate; or acyclovir stopped on a single negative swab.

The numbers worth carrying are these. Infection explains about one cholestatic infant in nine, cytomegalovirus is a third of those and bacteria nearly half. Endotoxin cholestasis is transcriptional and reversible. A catheter urine culture belongs early in the work-up. Congenital cytomegalovirus needs a positive sample within twenty-one days, or a positive newborn blood spot. Six months of valganciclovir improves hearing across both ears, seventy-seven versus sixty-four percent at two years. Virus positive atresia clears jaundice less often, fifteen versus fifty-two percent in one series, but it never delays the cholangiogram. American congenital syphilis reached 3,882 cases in 2023. And in neonatal liver failure, acyclovir at sixty milligrams per kilogram per day starts before the polymerase chain reaction result, because viral liver failure kills about two infants in three.

Part thirteen. What to do differently in clinic.

Let us return to the ward and apply the logic. The five-week-old with pale stools and a positive urine cytomegalovirus test should have biopsy and cholangiography on the usual schedule; meanwhile, retrieve the newborn blood spot to find out whether the infection was congenital. The three-week-old who is vomiting, has a direct bilirubin of three, and grows E. coli in the urine needs

antibiotics, lactose stopped, and a red cell galactose enzyme test or urine reducing substances. The ten-day-old with fever, an alanine aminotransferase of 2,400 and an international normalized ratio of three and a half, with no rash, needs acyclovir twenty milligrams per kilogram every eight hours now, blood and spinal fluid polymerase chain reaction, surface swabs including enterovirus testing, and early contact with a liver unit. The two-week-old with an enlarged liver and spleen, a snuffly nose and an untreated mother with a high nontreponemal titer has scenario one congenital syphilis and needs spinal fluid examination, long bone films, an eye examination and ten days of intravenous aqueous penicillin. And the twelve-day-old with symptomatic congenital cytomegalovirus and a small head has moderate to severe disease and should start valganciclovir, sixteen milligrams per kilogram twice daily for six months, with neutrophil monitoring.

The whole chapter can be held in two sentences. Bacterial infection switches the bile pump down and is cured by antibiotics, but it never closes the atresia question. Viral infection in the first weeks of life can destroy the liver within days, so in any sick neonate with a damaged liver, treat for herpes first and ask questions afterward.

Questions to test yourself.

Question one. Which transporters are down-regulated in endotoxin-induced cholestasis? Answer: the uptake transporters NTCP and the OATPs, and the export pumps BSEP and MRP2, while the escape pumps MRP3 and MRP4 are increased.

Question two. What share of infant cholestasis is due to infection, and which infection is most common? Answer: about eleven and a half percent, roughly one in nine; cytomegalovirus accounts for about a third of the infectious cases.

Question three. Why can a positive urine cytomegalovirus test at six weeks not prove congenital infection? Answer: because postnatal infection from breast milk or transfusion is possible after three weeks; congenital infection needs a positive sample within twenty-one days or a positive stored newborn blood spot.

Question four. What did six months of valganciclovir achieve compared with six weeks? Answer: better hearing across both ears, seventy-three versus fifty-seven percent at twelve months and seventy-seven versus sixty-four percent at twenty-four months, and better language scores at two years.

Question five. What is the significance of cytomegalovirus IgM in biliary atresia? Answer: it is associated with lower jaundice clearance after Kasai, fifteen versus fifty-two percent in one series, but a causal role is unproven and it must never delay the atresia work-up.

Question six. How is a cholestatic infant with congenital syphilis treated? Answer: as CDC scenario one, with aqueous penicillin G fifty thousand units per kilogram intravenously every twelve hours in the first week and every eight hours thereafter, for ten days.

Question seven. Which virus is the commonest identified viral cause of liver failure in infants under ninety days, and how lethal is viral liver failure? Answer: herpes simplex virus, about

thirteen percent of cases; viral liver failure carried a sixty-four percent risk of death without transplant.

Question eight. What happened in the 2022 to 2023 French echovirus 11 cluster? Answer: nine severe neonatal cases, mostly preterm and twins, with seven deaths from sepsis with hepatic failure.

End of chapter.